

UCSD MGTA 451: Business Analytics in Marketing, Finance and Operations

Marketing Module Syllabus

Professor [Kenneth C. Wilbur](#) and Instructional Assistant [Omri Shimoni](#)

[Marketing Module materials link](#)

Welcome to UC San Diego, and to MGTA 451! This class is a little unusual in that it is split into three intensive *modules* introducing analytics topics in marketing, operations and finance. Each module is led by a different professor and has its own syllabus. Each professor will grade every student in the class, and then the three module grades will each count as $\frac{1}{3}$ of the final assessment. This syllabus describes the marketing module only.

1. Marketing module learning objectives

Objective	Assessment
Understand and apply key concepts in marketing and customer value	Customer Value Discussion, Quiz 1
Interview customers	Customer Interview Exercise
Understand and apply key concepts in advertising measurement and causality	Advertising Measurement Discussion, Quiz 2
Understand and apply marketing mix model concepts	MMM Exercise

2. Preparation & Resources

Please do the following before we meet:

1. Fill out the pre-class survey at <https://forms.gle/zq7xp2Src8H3gs4Q6>
2. Read:
 - a. [Marketing and Customer Value](#), pages 1-36

Please do the following before Day 2:

3. Read:
 - a. [“Packaging Up Media Mix Modeling: An Introduction to Robyn’s Open-Source Approach”](#) by Runge, Skokan, Zhou (2024)
 - b. [“Proving Marketing Impact” by Think with Google](#)
 - c. [“Close Enough? A Large-Scale Exploration of Non-Experimental Approaches to Advertising Measurement”](#) by Gordon, Moakler, Zettelmeyer (2022)

All other required materials will be provided at the [Marketing Module materials link](#).

3. Assessments and Grading

The marketing module component will count for $\frac{1}{3}$ of your MGTA 451 grade. The marketing module grade will be determined by class contribution ($\frac{1}{3}$), day 1 activities ($\frac{1}{3}$), and day 2 activities ($\frac{1}{3}$). All grades will be curved and should be interpreted as relative, not absolute.

Class contribution will depend on productive contributions to class discussions, using Professor Nijs' 0-3 grading scale, and on in-class exercises on days 1 and 2. Quizzes will be on paper. Quiz questions will ask you to convey your clear understanding of primary concepts in readings and class discussions. You will not be allowed to access materials or devices during the quiz.

You are required to notify the professor and IA *in advance* if you will miss class due to a documented, university-sanctioned reason (e.g., illness with doctor's note, death in the family, etc.). In that case you can make up the quiz later as an oral exam.

4. Communication

We meet in person to facilitate two-way open-ended interactions. You should bring your questions, comments and observations to the class, to test your understanding, and to enable peer-to-peer learning. It is essential that you arrive prepared. We move fast, so there is no second chance to have a productive classroom experience.

We require Piazza for asynchronous communication, available [here](#). Main reasons:

- 1) Piazza enables students to answer each other's questions. Peer answers tend to be fast and accurate, and therefore encourage efficiency and fairness. Answers can inform other students, and help to ensure that everyone can reference the same information.
- 2) Piazza enables private messages to 'All Instructors' for personal situations such as illness. This is the place to notify us in advance if you have to miss class for a university-sanctioned reason. Messages must be addressed to All Instructors so the Professor & IA can coordinate and share records.

By default, any Piazza message should be visible to classmates, unless it contains personal or confidential information. You can post anonymously to classmates if you prefer, but we encourage use of names as the default. If you are posting confidential or private information, please post it to 'All Instructors' so the professor and IA can both access it.

We discourage use of emails and canvas/instructure messages, as they forego most Piazza benefits. We kindly ask you to note our communication preference, and in return we promise to answer Piazza messages faster than any other asynchronous medium. If you email us, we will ask you to kindly post your message to Piazza before we respond.

5. Welcome

Thank you for reading this document. We are excited to get to know all of you, and work with you in this course!

MGTA 451 FINANCE MODULE

SUMMER 2026 MSBA PROGRAM

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OFFICE HOURS by appointment
TEACHING ASSISTANT/TUTOR TBD
TA EMAIL [TBD](#)

CLASS SESSIONS According to Schedule
CLASSROOM 1E107

DESCRIPTION

This short course is designed to provide strong foundations in the fundamental concepts of Finance. We will discuss key concepts, techniques and calculations, as well as applications to real world situations. The course is structured as a combination of lectures, in-class exercises, discussions, and assignments.

There are no prerequisites to this course.

OBJECTIVES

By the end of the course, students are expected to possess the skills for:

- *using time value of money concepts*
- *applying investment decision rules*
- *evaluating and choosing projects*
- *understanding the basics of asset valuation*
- *effectively comparing risk and return tradeoffs in a wide variety of financial dilemmas*
- *recognizing how diversification leads to better portfolio outcomes*

MATERIALS

Required

Course materials: lectures and handouts will be available on Canvas.

COURSE SCHEDULE

We will be covering a substantial amount of material in a relatively short time. The course outline provides a general framework that may be adjusted as necessary to accommodate our progress.

Session	Class Topic & Activities
1	Introduction: The Field of Finance Interest Rates Role and Actors on Financial Markets, Time Value of Money Risk and Return Discussion: Retirement Crisis in the Developed World Problems and Examples
2	Arbitrage and the Law of One Price Diversification Asset Valuation Principles Investment Decision Rules: The Concept of Net Present Value Market Efficiency Discussion: Stock Market Bubbles

ASSIGNMENTS

Rady Honor Code, Class Policies on Academic Integrity and the Use of AI Tools apply to all assignments. Please see these under the Course Policies section of the syllabus.

Homework Assignment

Group Work.

Two sets of homework questions and problems covering the first lecture will be assigned to students. The deadline for the first homework is September 11, and for the second homework September 15.

Teams share the workload as they see fit. I recommend that each team member solves the problems, and team members discuss the solutions to arrive at the correct answers. Team members receive the same score unless they notify the instructor and TA via email that one of the members did not sufficiently contribute to the assignment. In this case, the grade will be reduced for the free-riding student.

The homework assignments comprise 5 gradepoints each, 11 gradepoints total.

Take Home Assignment and Oral Examination

Independent work.

Upon completion of the Finance Module of MGTA 451 students will complete and submit a take-home assignment. The submission deadline is September 21, 2025. The take home examination will comprise 6 gradepoints.

The take home examination will be followed by an oral examination at the end of the course. The date of the oral examination will be posted on Canvas. The oral examination also counts for 6 gradepoints.

Class Participation

Independent work.

Class participation is an important way to enhance the learning experience for everyone. Participation can take the form of asking relevant questions, making comments, or citing an experience or viewpoint. Class participation also helps students to gain confidence in asserting their positions in unfamiliar environments and listening to others' perspectives.

11.33 gradepoints are awarded for class participation. This grade will reflect my assessment of both the quantity and quality of the individual's contribution to the learning experience for everyone in the class.

Class participation will be recorded according to the following guidelines in each of the two classes:

Excellent	Insightful contributions in each class demonstrating thorough preparedness, critical thinking and a good grasp of the material
Satisfactory	Fewer, but relevant contributions reflecting adequate preparedness and understanding of the material
Needs Improvement	Class attendance and participation in iClicker surveys without extra contributions
Unsatisfactory	Shortcomings in class attendance and participation in iClicker surveys

GRADING

Exams and Assignments	Points [or percentage]
Homework Assignments (2 x 5.5p)	10
Take-Home Examination	6
Oral Examination	6
Class participation	11.33
Total	33.33

COURSE POLICIES

Finance is a fascinating field for researchers and practitioners alike. Class discussions give ample opportunity to learn from each other's varied experiences and perspectives. We strive to practice meaningful and constructive dialogue in an atmosphere of mutual respect, willingness to listen, and tolerance of opposing points of view.

Communication with the Instructor and the Teaching Assistant

We aim to create an inclusive learning environment that supports diversity of thoughts, perspectives and experiences, and honors identities (including race, gender, class, sexuality, religion, ability, etc.). Students are encouraged to make suggestions to improve focus on diversity and inclusion and bring sensitive issues to my attention via email. I will respect requests to keep anonymity.

Contributions that are of interest to the entire class should be posted to Piazza. Anonymous contributions are not permitted on the class Piazza.

Individual questions (say about grading, or students' unique circumstances, etc.) should be sent to the faculty and the TA via email.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The Rady Honor Code applies to all assignments in this course.

No student shall engage in any activity that involves attempting to receive a grade by means other than honest effort, for example:

- **no student shall knowingly procure, provide, or accept any unauthorized material that contains questions or answers to any examination or assignment to be given at a subsequent time,**
- **no student shall complete, in part or in total, any examination or assignment for another person,**
- **no student shall knowingly allow any examination or assignment to be completed, in part or in total, for himself or herself by another person,**
- **no student shall plagiarize or copy the work of another person and submit it as his or her own work,**
- **no student shall employ aids excluded by the instructor in undertaking coursework or in completing any exam or assignment,**
- **no student shall alter graded class assignments or examinations and then resubmit them for re-grading, and**
- **no student shall submit substantially the same material in more than one course without prior authorization.**

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

CLASS POLICIES ON THE USE OF ARTIFICIAL INTELLIGENCE (AI) TOOLS LIKE CHATGPT, BARD, BING CHAT, CLAUDE, ETC.

AI tools such as ChatGPT are permitted in this class for the purposes of enhancing learning.

However, it is the responsibility of each student to ensure that their use of AI tools does not constitute academic misconduct, such as plagiarism or cheating.

In other words students are responsible for

- *verifying the accuracy of content generated by AI tools if they use this content in graded assignments, such as the team project, or discussion board contributions;*
- *referencing such content to its proper AI source;*
- *employing AI tools in a responsible and ethical manner; and*

- *abiding by the guidelines set forth by the instructor.*

Any violation of these guidelines will result in disciplinary action.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the instructor and OSD liaison.

Students requesting accommodations for this course due to a disability must provide a current Authorization for Accommodation (AFA) letter (paper or electronic) issued by the Office for Students with Disabilities (<https://osd.ucsd.edu/portal/tutorial.html>). Students are required to discuss accommodation arrangements with instructors and OSD liaisons in the department **72 business hours in advance** of any exams or assignments. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

NONDISCRIMINATION POLICY STATEMENT

The University of California, in accordance with applicable federal and state laws and university policies, does not discriminate on the basis of race, color, national origin, religion, sex, gender, gender identity, gender expression, pregnancy (including pregnancy, childbirth, and medical conditions related to pregnancy or childbirth), physical or mental disability, medical condition, genetic information, ancestry, marital status, age, sexual orientation, citizenship, or service in the uniformed services (including membership, application for membership, performance of service, application for service, or obligation for service in the uniformed services). The university also prohibits harassment based on these protected categories, including sexual harassment, as well as sexual assault, domestic violence, dating violence, and stalking. The nondiscrimination policy covers admission, access, and treatment in university programs and activities.

TITLE IX

The Office for the Prevention of Harassment & Discrimination (OPHD) provides assistance to students, faculty, and staff regarding reports of bias, harassment, and discrimination. OPHD is the UC San Diego Title IX office. Title IX of the Education Amendments of 1972 is the federal law that prohibits sex discrimination in educational institutions that are recipients of federal funds. Rady students have the right to an educational environment that is free from harassment and discrimination.

You can make a complaint of harassment or discrimination — or simply make an appointment to find out more information — by contacting OPHD.

- (858) 534-8298
- ophd@ucsd.edu

- [Overview for Students](#) webpage

Students may feel more comfortable discussing their particular concern with a trusted employee. This may be a Rady student affairs staff member, a department Chair, a faculty member or other University official. These individuals have an obligation to report incidents of sexual violence and sexual harassment to OPHD. This does not necessarily mean that a formal complaint will be filed.

If you find yourself in an uncomfortable situation, ask for help. The Rady School of Management is committed to upholding University policies regarding nondiscrimination, sexual violence and sexual harassment.

HEALTH AND WELL-BEING

Throughout your time at UC San Diego, you may experience a range of issues that can negatively impact your learning. These may include physical illness, housing or food insecurity, strained relationships, loss of motivation, depression, anxiety, high levels of stress, alcohol and drug problems, feeling down, interpersonal or sexual violence, or grief.

These concerns or stressful events may lead to diminished academic performance and affect your ability to participate in day-to-day activities. If there are issues related to coursework that are a source of particular stress or challenge, please speak with me, [Professor's Name], so that I am able to support you. UC San Diego provides a number of resources to all enrolled students, including:

- Counseling and Psychological Services (858-534-3755 | caps.ucsd.edu)
- Student Health Services (858-534-3300 | studenthealth.ucsd.edu)
- CARE at the Sexual Assault Resource Center (858-534-5793 | care.ucsd.edu)
- The Hub Basic Needs Center (858-246-2632 | basicneeds.ucsd.edu)

OPERATIONS MODULE

Instructor: Shirin Shahsavand
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Tel: 619-800-6251

Teaching Assistant: Divij Chokhani
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Rules & Grading Scheme for Operations Component

Operations component has a weight of 1/3 of your overall class grade

Component	Percentage
Homework: Work in assigned groups but submit your own answers in Canvas	30%
Exam (written and oral): Must work individually – discussion of the exam with others is <u>not permitted</u> .	70%

Late assignments on Canvas **will not** be accepted! Be sure to submit on time.

Operations Session 1	Operations Session 2
• Introduction to the operations function of a business • A taxonomy of operations management • Alignment of competitive and operations strategy • Process analysis – Key process-flow measures – Inventory buildup diagrams – Little's Law – Inventory turns • In-class problems – Inventory build-up analysis <u>Homework</u> Submit your solution in Canvas prior to the listed deadline. ○	• Structuring operations • Product-process matrix • Introduction to capacity and processes • Process analysis – Capacity Analysis – Flow Time Analysis • In-class problems – State Automobile License Renewal Case <u>Individual Exam</u> Submit your solution in Canvas prior to the listed deadline